

year, will also be packed with lots of technology like an organic LED (OLED) display, improved position tracking with nine LEDs and a refresh rate of 120Hz.

Fove VR, another headset expected to start shipping later this year, also packs in some interesting technology such as eye-tracking. Like Oculus, Fove is another Kickstarter success story. It is partly funded by Samsung Ventures. Fove VR can track subtle eye movements. This makes it possible to make eye contact with virtual characters, to change your view by simply shifting your eyes, and to aim better and shoot in games like archery. The view in front of you also blurs or sharpens depending on how you focus, making your virtual experience more natural. Sometimes, unnatural and frequent movement of the head to change your view of the virtual environment tends to cause vertigo and nausea. Eye-tracking allows you to change views more naturally, thereby overcoming these problems, too.

Microsoft's VR cum AR offering, HoloLens, is way beyond this price range. With development editions shipping now, the price estimates are around US\$ 3000. However, it is a different genre in itself as it populates your virtual world with high-definition holograms. The environment is more immersive, the characters are claimed to be almost life-like, and you can move about and interact with them. HoloLens is not tethered to a PC or a mobile phone, as a complete Windows 10 system is built into the battery-powered headset. Developers can build holographic apps for HoloLens using the Universal Windows Platform.

An open source ecosystem for VR

When you think of taking a technology to more people, it is not really possible without an open source ecosystem. Open Source Virtual Reality (OSVR) is an open source



Microsoft HoloLens, the first fully-untethered holographic computer, enables users to interact with high-definition holograms

ecosystem that aims to unify VR and AR technologies, so that headsets, accessories and apps can be used across platforms. The OSVR software platform drives this kind of end-to-end compatibility through a single VR ecosystem. It hopes to overcome the problems that developers face due to hardware and platform fragmentation. It aims to create the perfect scenario wherein developers can focus on their work without worrying about hardware or content support. OSVR is backed by big names like Leap Motion, Intel and Gearbox Software.

Razer recently launched an OSVR Hacker Development Kit, which includes the complete designs and schematics required to build your own VR headset. The design is completely open and can be expanded, modified or upgraded. You can, for example, upgrade the faceplate or add a position tracker later to improve on the equipment as and when you wish. This serves all the purposes of sustainability and accessibility by enabling reuse of components and cross-platform compatibility.

Of what use is an affordable headset

However affordable it may be, a headset is worthless if enough VR content is not made available to the masses. It will not be wrong to say that the real VR revolution has been on the content side. In the recent

past, there has been a splurge in the number of VR apps available on Google Play and Apple App Store, as well as the number of 3D video channels on YouTube. The most pleasant surprise, however, is the VR content uploaded by The New York Times. In their words, they put you in the centre of the stories only they can tell!

All this is possible thanks to the VR development platforms and 360-degree cameras available today for VR content development. Google's Jump, for example, is an entire ecosystem for VR content development. It consists of a camera rig, software that assembles the footage seamlessly together and a player.

Google, in association with GoPro, has designed and developed a 360-degree, 16-camera rig for this purpose. However, it is not necessary that only the GoPro camera will work with the Jump ecosystem. Theoretically, you can use any camera. Or, you can build your own camera rig using Google's specifications, which cover everything from the size and shape of the rig to the placement of cameras and their FOV.

As the next step, the back-end software converts the 16 raw video feeds from the cameras into VR video in stereoscopic 3D by combining computational photography and computer vision. Basically, it recreates the scene by seamlessly combining the camera footage with thousands of computer-generated, intermediate viewpoints. As for the player, Google simply added VR content as an option on YouTube. What better way for it to reach millions of people every day!

Facebook has also designed a high-end video capture system called Surround 360, the reference designs of which will soon be released as an open source